



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Manohar Vinay M
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Outline

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- Methodology
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Executive Summary

- Historical data on falcon 9 launches is analysed to identify patterns and factors influencing first stage landing success.
- Machine learning models are utilised to predict landing outcomes based on variables like payload flight trajectory and weather conditions.
- Data preprocessing steps ensure accuracy and reliability, including feature engineering and validation techniques.
- The models demonstrate a high accuracy in predicting first stage landing success.
- Key factors contributing to successful landing are identified, providing actionable insights.
- The findings enable cost estimations for launches and facilitate competitive strategies for aerospace industry players.

Introduction

- SpaceX's falcon 9 rockets achieve cost efficiency by reusing their first stage, reducing launch costs from \$165 million to \$62 million per flight This economic advantage is a significant factor in the aerospace industry.
- Predicting whether the falcon 9's first stage will successfully land is crucial for understanding launch costs and planning competitive strategies in the market.
- The ability to accurately predict first stage landing success is essential for cost estimation risk assessment, and optimization of resources for rocket launches.
- The findings can guide rival companies in bidding for contracts, improve decision making and promote further cost efficiency in the space launch market.
- Success in this endeavour can contribute to advancements in reducible rocket technology and foster competition in the aerospace industry.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data is collected by extracting historical falcon 9 launch records including various metrics from publicly available resources and databases
- Perform data wrangling
 - The data was Cleaned, transformed and prepared for analysis using techniques like feature engineering and normalisation.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - This includes data split, model training, hyperparameter tuning and performance evaluation

Data Collection

- Falcon 9 launch courts were extracted from publicly available sources such as Wikipedia, using web scrapping tools like beautiful soup.
- SpaceX's API was utilised to gather detailed launch data, including payload, orbit type and machine outcomes.
- Additional data was sourced from online databases and repositories to ensure comprehensive coverage.
- The collected data included various formats such as Json and HTML tables, which were later converted into Panda's data frames for analysis.
- We collected data was cross verified for accuracy and consistency to ensure reliability for further analysis.
- The dataset spans multiple years of Falcon nine launches, providing a robust foundation for predictive modelling

Data Wrangling

- Identified and addressed missing data points to ensure data set completeness and reliability.
- Removed duplicates, corrected inconsistencies and standardised formats for accurate analysis.
- Created new variables and transformed existing ones to enhance predictive modelling.
- Merge data sets for multiple sources such as Apis and web scrapping into unified structure.
- Converted categorical variables into numerical formats using techniques like one hot encoding.
- [Data-Science-Capstone-Project/labs-jupyter-spacex-Data wrangling.ipynb at main · vinay124-tech/Data-Science-Capstone-Project](#)

EDA with Data Visualization

- Created scatter plots to analyse relationships between variables such as payload mass and landing success.
- Used bar charts to visualise categorical data, like the frequency of successful landings across different launch sites.
- Generated heat maps to identify correlations between numerical features, aiding in feature selection for modelling.
- Built dashboards using Plotly Dash to explore data dynamically and uncover patterns.
- Visualised launch locations and proximity using folium, providing insights to spatial factors affecting outcomes.
- [Data-Science-Capstone-Project/EDA with Visualisation.ipynb at main · vinay124-tech/Data-Science-Capstone-Project](#)

EDA with SQL

- Used SQL queries to extract relevant subsets of data from the database for analysis.
- Applied SQL Commands like ORDER BY and WHERE to sort and filter database on specific conditions.
- Utilised functions like SUM, AVG and COUNT to calculate key statistics and summarise the data.
- Combined data from multiple tables using SQL joins to create a comprehensive data set for analysis.
- Used group by to analyse data trends and patterns across different categories of the data.
- Leveraged SQL queries to uncover meaningful insights, such as correlations and anomalies guiding further analysis
- [Data-Science-Capstone-Project/jupyter-labs-eda-sql-coursera_sqllite.ipynb at main · vinay124-tech/Data-Science-Capstone-Project](https://github.com/vinay124-tech/Data-Science-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb)

Build an Interactive Map with Folium

- Markers were used to pinpoint specific locations such as spacex launch site, enabling precisely visualisation of key spatial points of interest.
- Circles were added to represent areas of influence or proximity around locations, aiding in the analysis of special relationships and coverage.
- Alliance incorporated to connect points illustrating trajectories or routes for launches to understand spatial connexions.
- Clusters are used for dynamic grouping of nearby markers to declutter dense areas, improving map readability and visual clarity.
- Interactive elements like pop ups and tool pits are attached to map objects, provide additional details and enhance user engagement.
- [Data-Science-Capstone-Project/Folium lab.ipynb at main · vinay124-tech/Data-Science-Capstone-Project](#)

Build a Dashboard with Plotly Dash

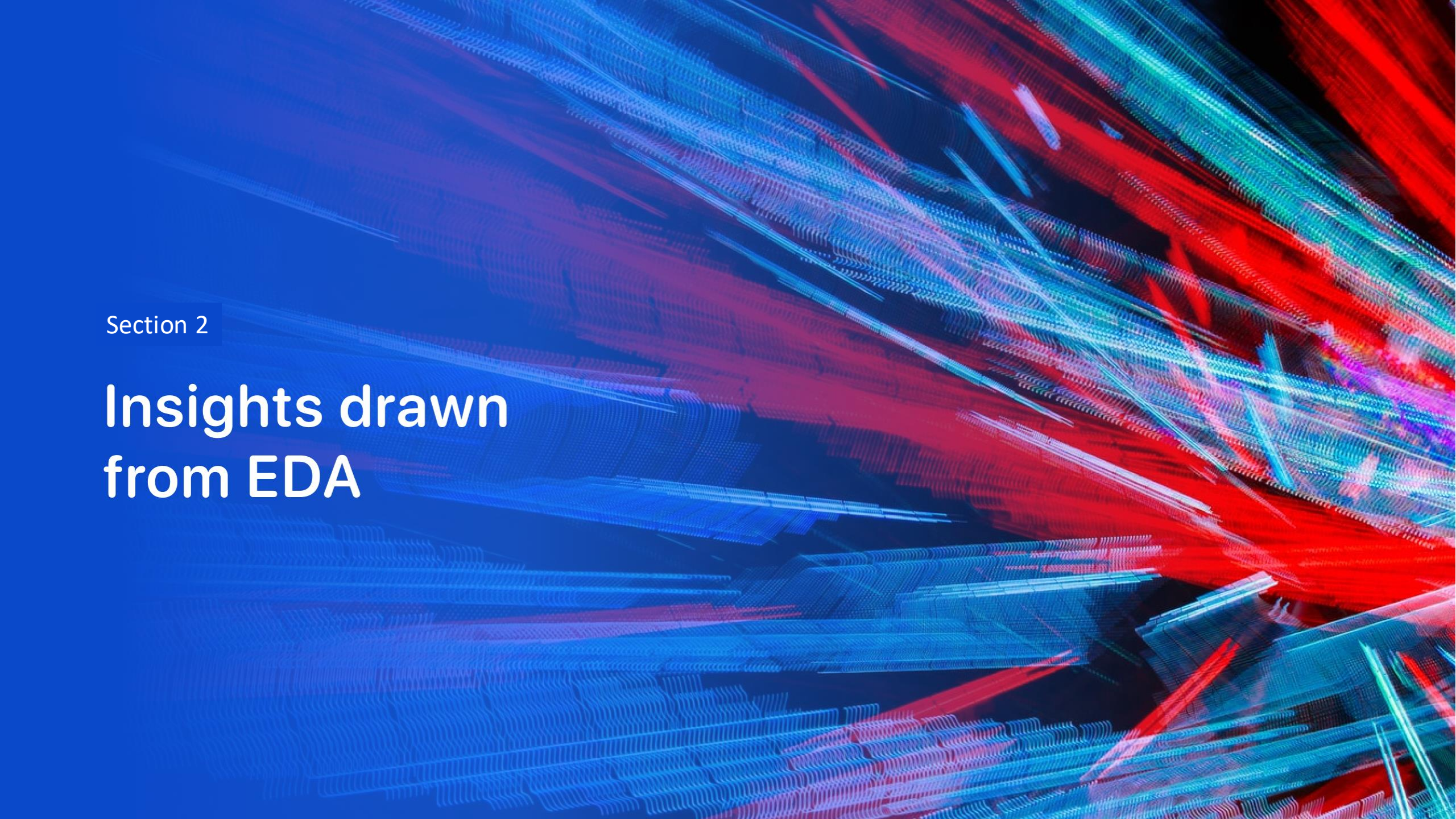
- Pie charts displays the success rate of launches for all sites are selected site. This helps users to quickly understand the proportion of successful verses failed launches.
- Scatter plots visualises the correlation between payload mass and launch site success, providing insights into half payload effects outcomes.
- Drop down menu allows users to filter the data by launch site, enabling focused analysis of specific locations.
- Range slider lets the users to select a payload mass range, dynamically updating these scatter plots to explore payload related trends.
- Interactive updates were implemented like callback functions to ensure that visualisations respond dynamically to user inputs, enhancing interactivity and Usability.
- So these elements are added to make the dashboard intuitive, interactive and effective finalising spaceX, Falcon 9 launch data.

Predictive Analysis (Classification)

- Data splitting divided the data set into training and testing subsets to ensure unbiased evaluation of models performance.
- Trained multiple classification algorithms such as logistic regression and decision trees and random forest to predict falcon 9 first stage landing success.
- Hyper parameter tuning optimised model parameters using grid search to enhance accuracy and prevent overfitting.
- Model evaluation assessed models using metrics like accuracy, precision, recall and F1 score to identify the best performing algorithm
- Selected the best model with the highest evaluation metrics and deployed it for predictive analysis.
- [Data-Science-Capstone-Project/SpaceX_Machine Learning Prediction_Part_5.ipynb at main · vinay124-tech/Data-Science-Capstone-Project](#)

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

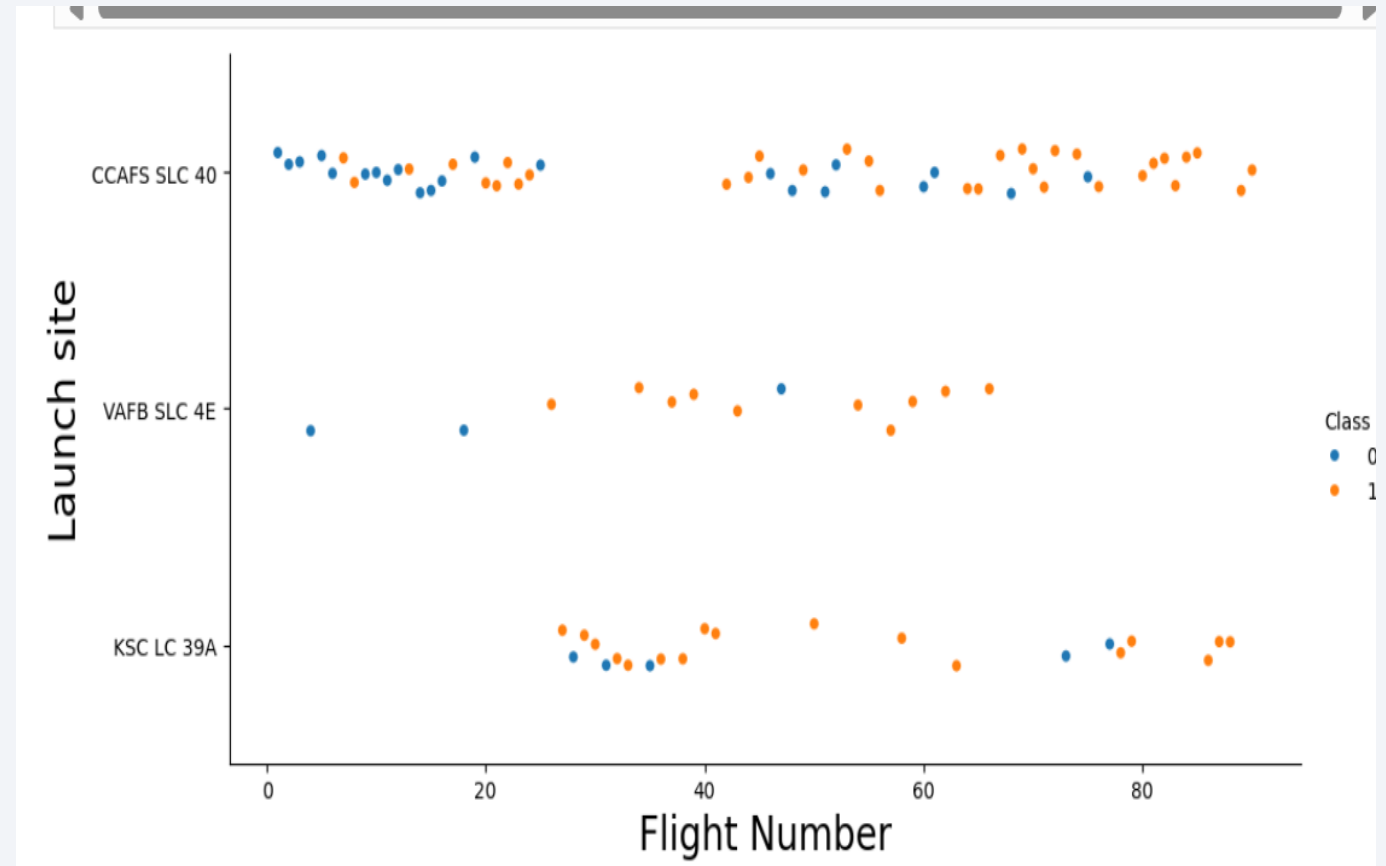
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

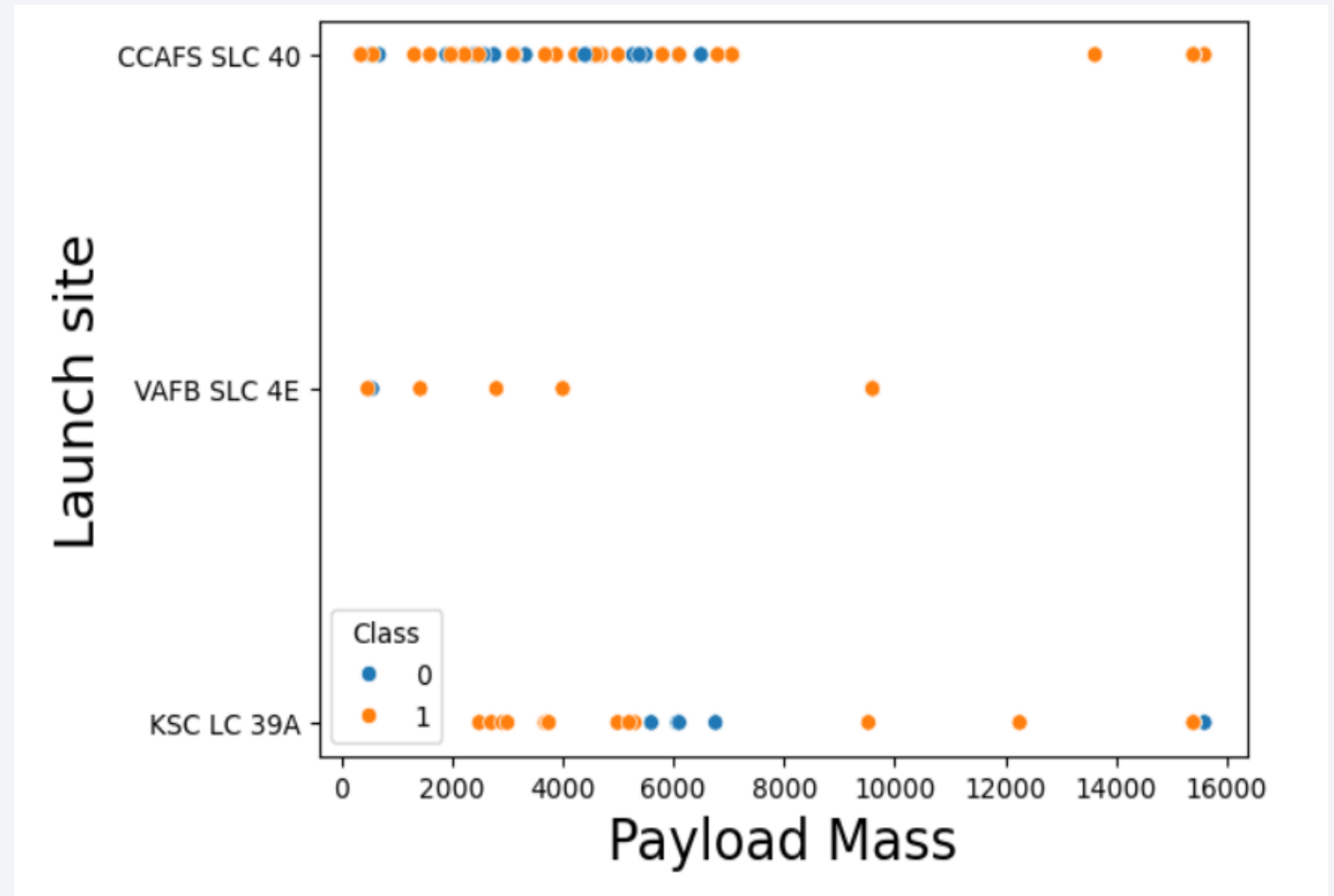
Flight Number vs. Launch Site

- The scatter plot shows the number of successful and failed flights at three different launch sites.
- CCAFS SLC 40 has more launch and success rate than the remaining two launch sites



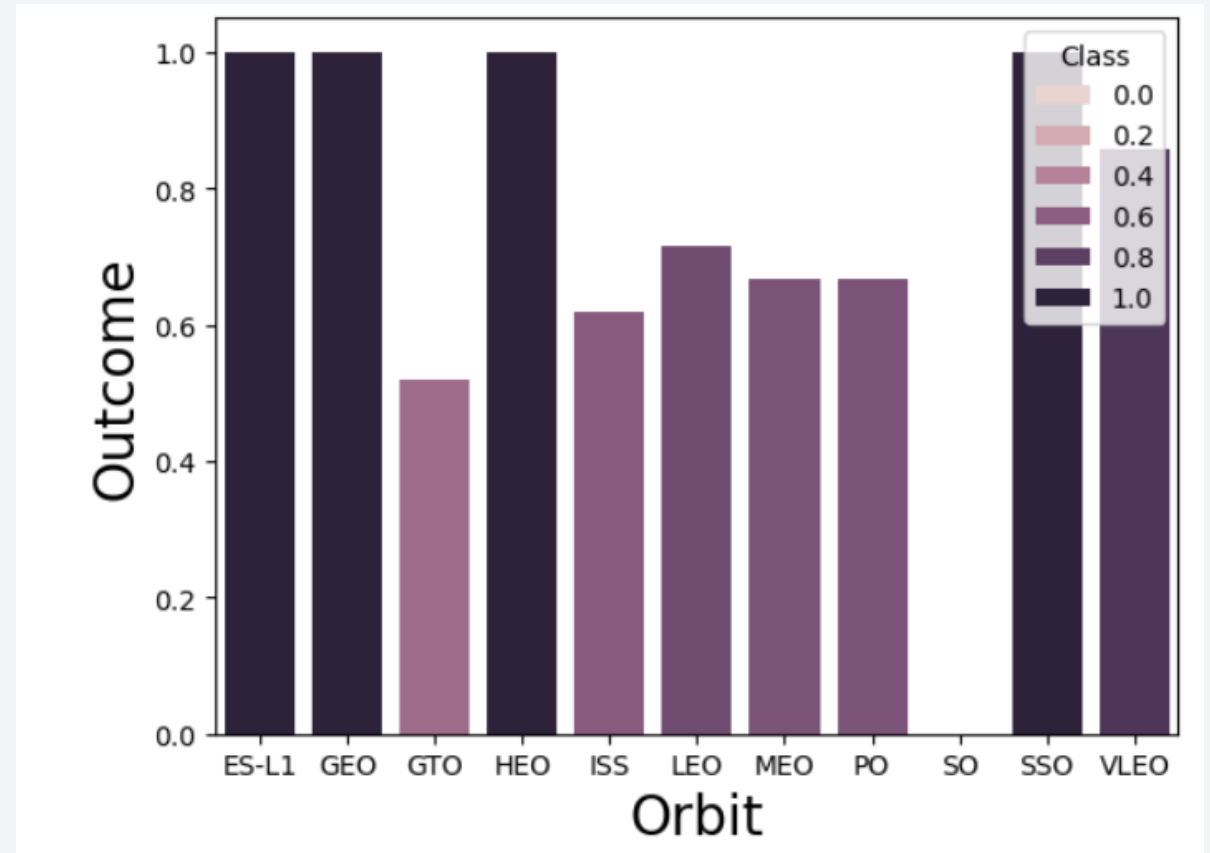
Payload vs. Launch Site

- This scatter plot of Launch sites vs Payload Mass shows the relation between payload mass and launch sites relationship.
- It can be observed that VAFB-SLC launch site has not launches any rockets with payload mass greater than 10000.



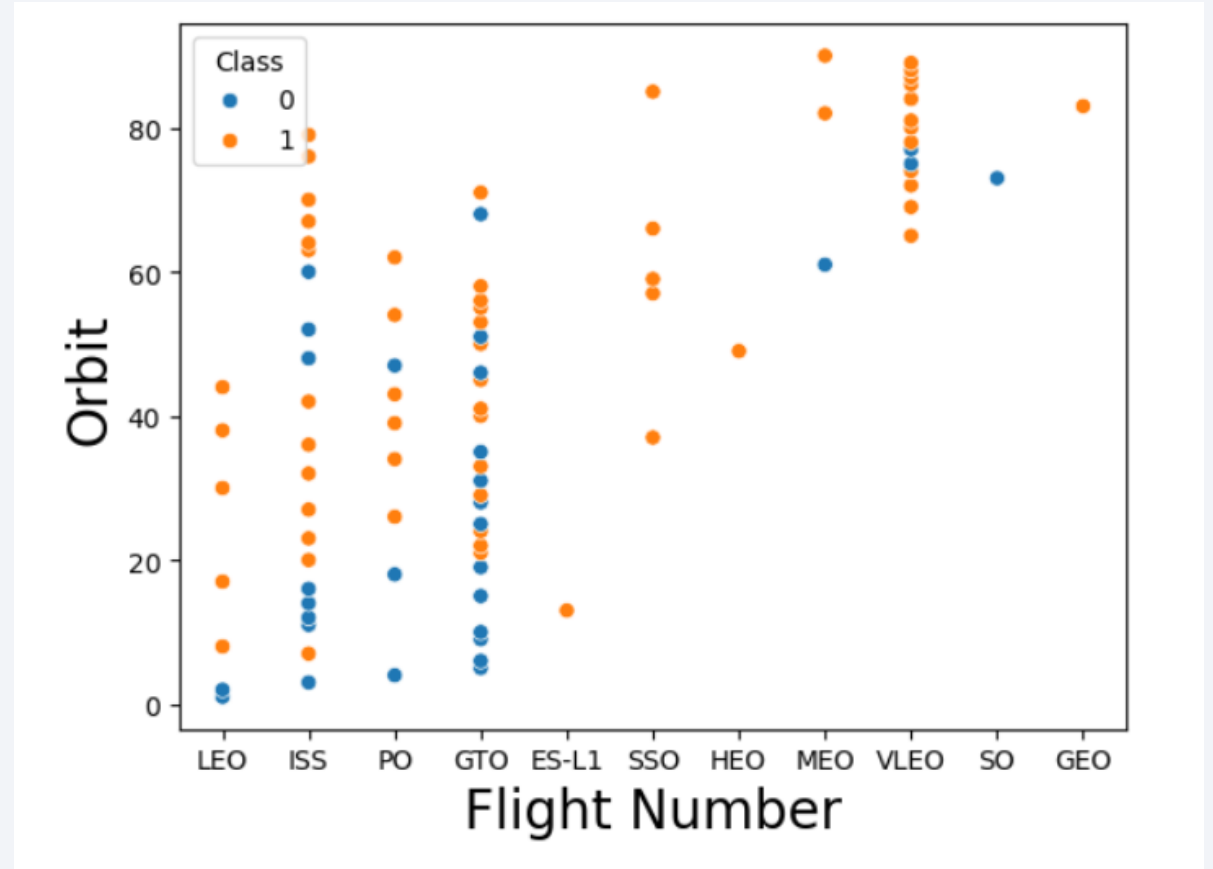
Success Rate vs. Orbit Type

- Bar chart showing Success rate at different orbits.
- Some Orbits like ES-L1, GEO, HEO, SSO has significant amount of success rate compared to other Orbits.



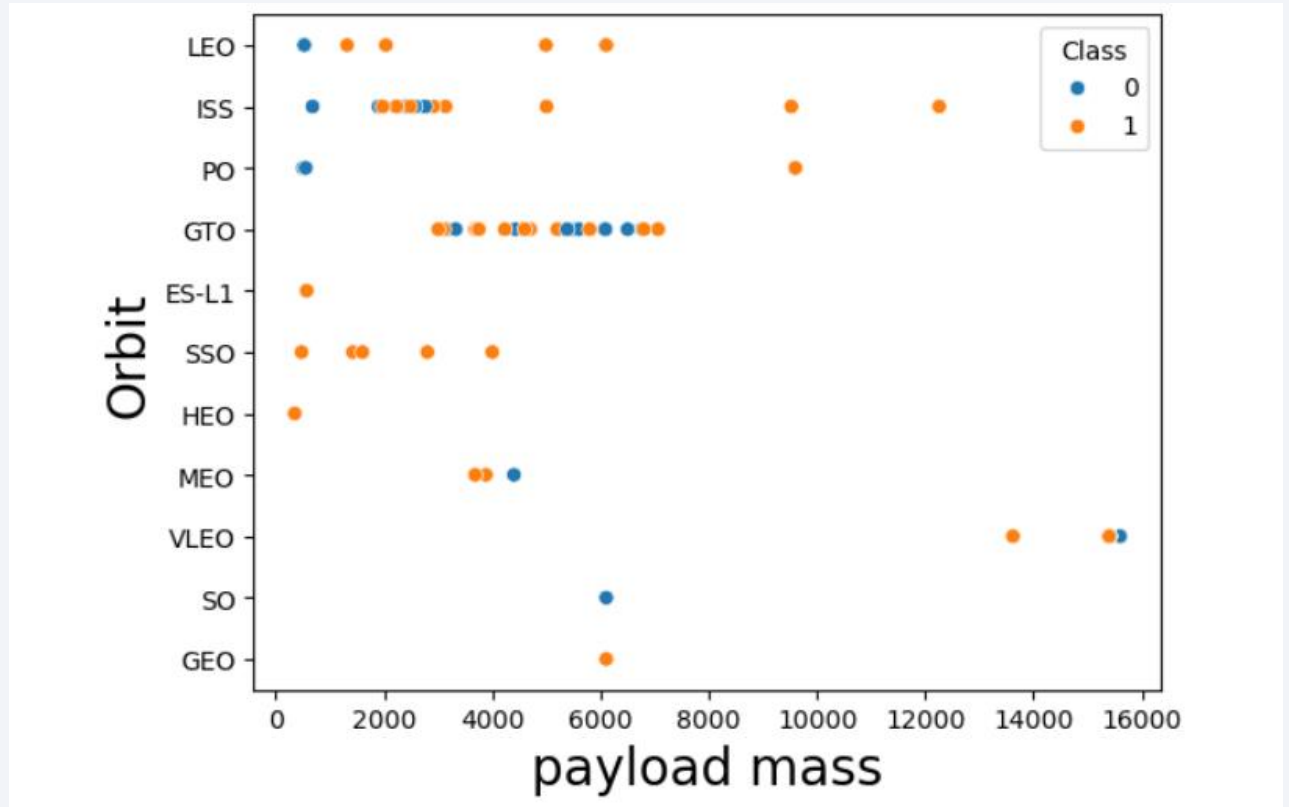
Flight Number vs. Orbit Type

- This scatterplot shows the number of flight launches per orbit.
- We can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in GTO orbit, there appears to be no relationship between flight number and the success.



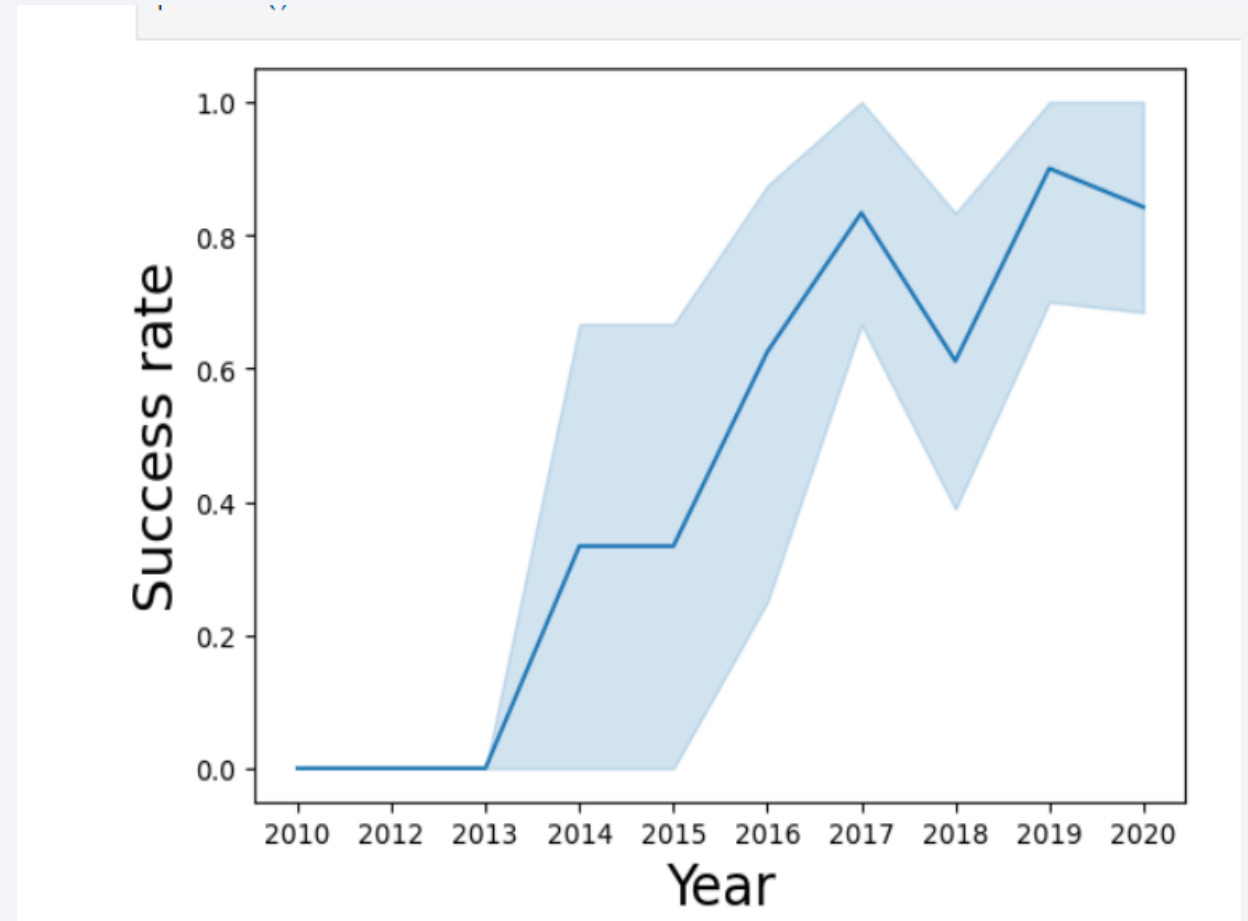
Payload vs. Orbit Type

- This scattered plot shows us that with heavy payloads, the successful landing, or positive landing rate, are more for polar, LEO and ISS orbits.
- However, for GTO, it's difficult to differentiate between successful and unsuccessful landings, as both outcomes are present.



Launch Success Yearly Trend

- Here is a line chart showing yearly average success rate.
- We can observe that the success rate since 2013 kept increasing till 2020.
- This is a best proof to say Falcon 9 is doing great and kept improving over years



All Launch Site Names

- Here is an SQL to get all the unique launch site names from the data frame It shows four launch sites of spaceX's falcon 9 rocket.

```
In [10]: %sql select distinct "Launch_Site" from SPACEXTABLE
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[10]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- Here is the SQL to present the launch sites beginning with CCA. And the first five results are shown here.

```
In [11]: %sql select * from SPACEXTABLE where "Launch_Site" like "CCA%" limit 5
```

```
* sqlite:///my_data1.db  
Done.
```

Out[11]:	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Sql query to display the total payload mass carried by boosters, launched by NASA (CRS)
- We can see there is a total of 45,496 kgs of payload, which was taken over by the NASA (CRS)

```
In [12]: %sql select SUM(PAYLOAD_MASS__KG_) as total_NASA_CRS_payload from SPACEXTABLE where Customer = "NASA (CRS)"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[12]: total_NASA_CRS_payload
```

```
45596
```


Average Payload Mass by F9 v1.1

- SQL query to calculate the average payload mass carried by booster version F9 v1.1
- Results show the Average payload mass carried by the booster F9 v1.1 is 2928.4 kg.

```
In [13]: %sql select AVG(PAYLOAD_MASS__KG_) as avg_F9_payload from SPACEXTABLE where Booster_Version = "F9 v1.1"
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[13]: avg_F9_payload  
          2928.4
```

First Successful Ground Landing Date

- This SQL query shows the date of the first successful landing outcome on ground pad
- The date is 22 December 2015.

```
In [14]: %sql select MIN(Date) as successful_groundpad_landing from SPACEXTABLE where Landing_Outcome = "Success (ground pad)"
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[14]: successful_groundpad_landing
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- This SQL query lists the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

In [15]: `%sql select Booster_Version, PAYLOAD_MASS_KG_, Landing_Outcome from SPACEXTABLE where PAYLOAD_MASS_KG_ between 4000 AND 6000`

* sqlite:///my_data1.db
Done.

Out[15]:

Booster_Version	PAYLOAD_MASS_KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Booster_Version	PAYLOAD_MASS_KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Total Number of Successful and Failure Mission Outcomes

- This SQL query gives us the total number of successful and failure mission outcomes
- Out of 101 Total missions only 3 were failed.

```
In [16]: %%sql SELECT
          COUNT(*) AS total_missions,
          SUM(CASE WHEN Mission_Outcome = 'Success' THEN 1 ELSE 0 END) AS successful_missions,
          SUM(CASE WHEN Mission_Outcome != 'Success' THEN 1 ELSE 0 END) AS failed_missions
          FROM
          SPACEXTABLE;
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[16]:
```

total_missions	successful_missions	failed_missions
101	98	3

Boosters Carried Maximum Payload

- This SQL query lists the names of the booster which have carried the maximum payload mass
- There are 12 boosters in total.

```
In [17]: %sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS__KG_ = (select MAX(PAYLOAD_MASS__KG_) from SPACEXTABLE);

* sqlite:///my_data1.db
Done.

Out[17]: Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7
```

2015 Launch Records

- Here is an SQL query to list the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015.
- We have 2 failed missions in the months of January and April.

In [24]: `%sql select substr(Date, 6,2) as month, Booster_Version, Launch_Site, Landing_Outcome from SPACEXTABLE where Landing_Outcome`

* sqlite:///my_data1.db

Done.

Out[24]:

month	Booster_Version	Launch_Site	Landing_Outcome
-------	-----------------	-------------	-----------------

01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
----	---------------	-------------	----------------------

04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)
----	---------------	-------------	----------------------

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Here is an SQL query to rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
In [31]: %sql select Landing_Outcome, count(Landing_Outcome) as outcome_count from SPACEXTABLE where Date between '2010-06-04' and '
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[31]:
```

Landing_Outcome	outcome_count
Precluded (drone ship)	1
Failure (parachute)	2
Uncontrolled (ocean)	2
Controlled (ocean)	3
Success (ground pad)	3
Failure (drone ship)	5
Success (drone ship)	5
No attempt	10

Landing_Outcome	outcome_count
Precluded (drone ship)	1
Failure (parachute)	2
Uncontrolled (ocean)	2
Controlled (ocean)	3
Success (ground pad)	3
Failure (drone ship)	5
Success (drone ship)	5
No attempt	10

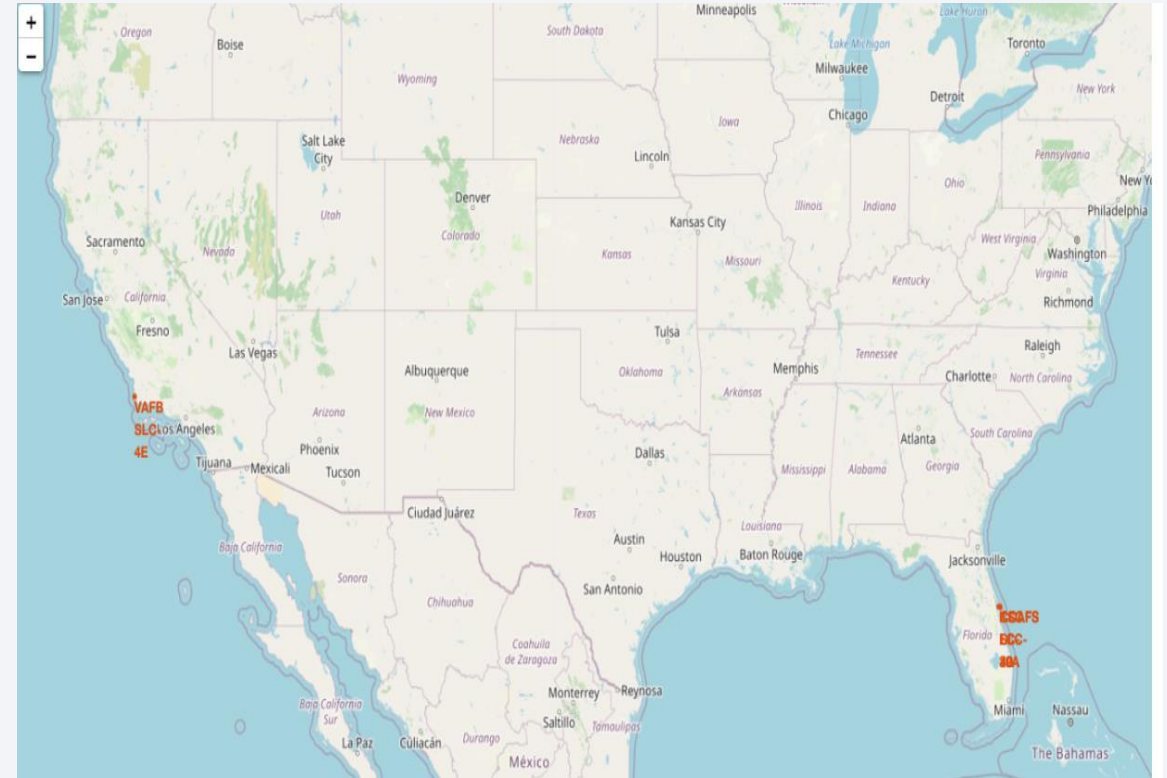
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite image of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The lights are concentrated in the lower right portion of the image, following the curve of the Earth's horizon. The overall composition suggests a global or space-based perspective.

Section 3

Launch Sites Proximities Analysis

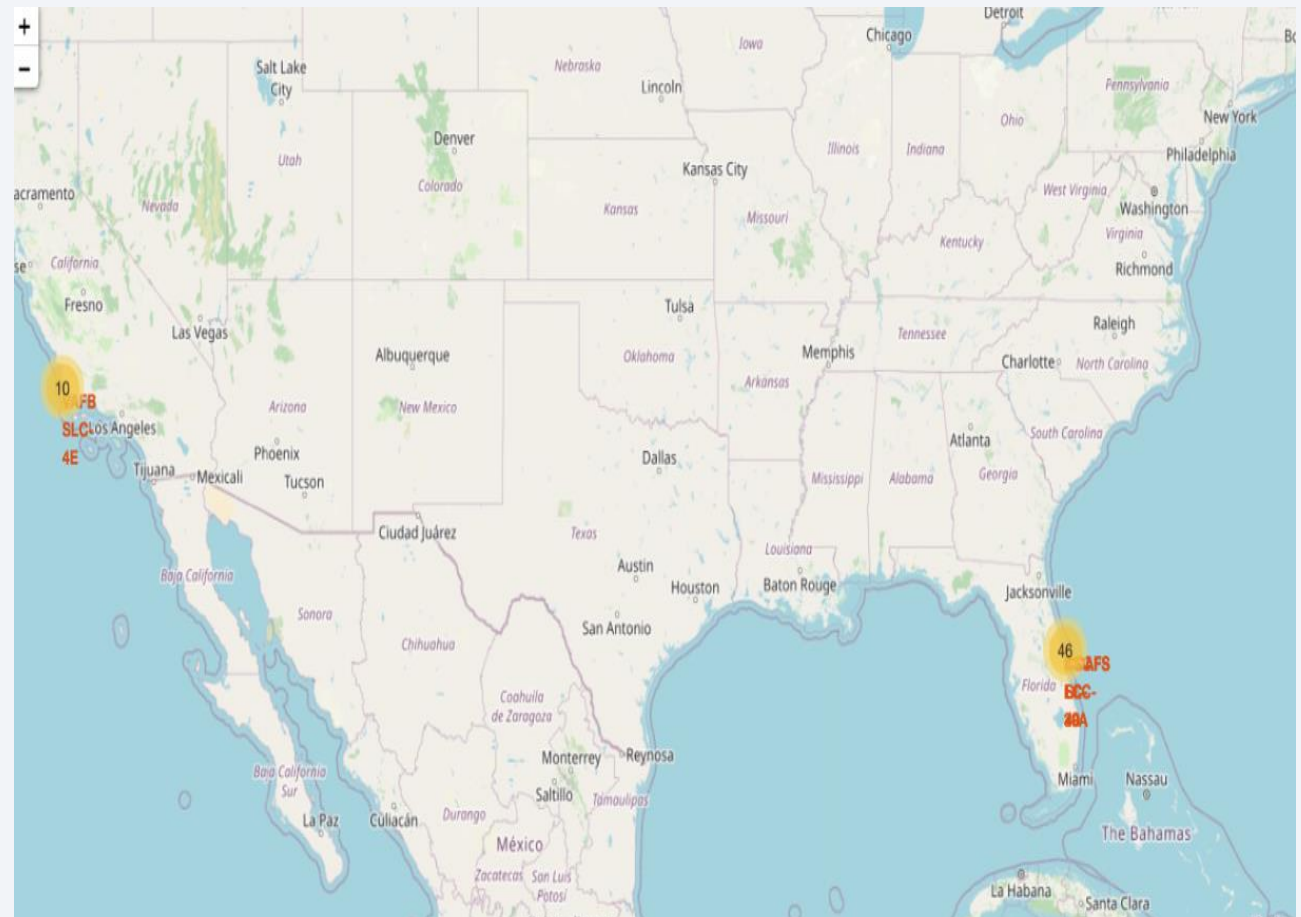
Folium Map Showing all the Launch sites

- Not all launch sites are closed to the equator For example, canary space centre and Cape Canaveral are located in Florida, which is relatively close to the equator compared to vandenburg Air Force base in california.
- All launch sites are indeed very close to the coast. This is a strategic choice to ensure safety during launches, as rockets can be directed over the ocean minimising the risks to populated areas.



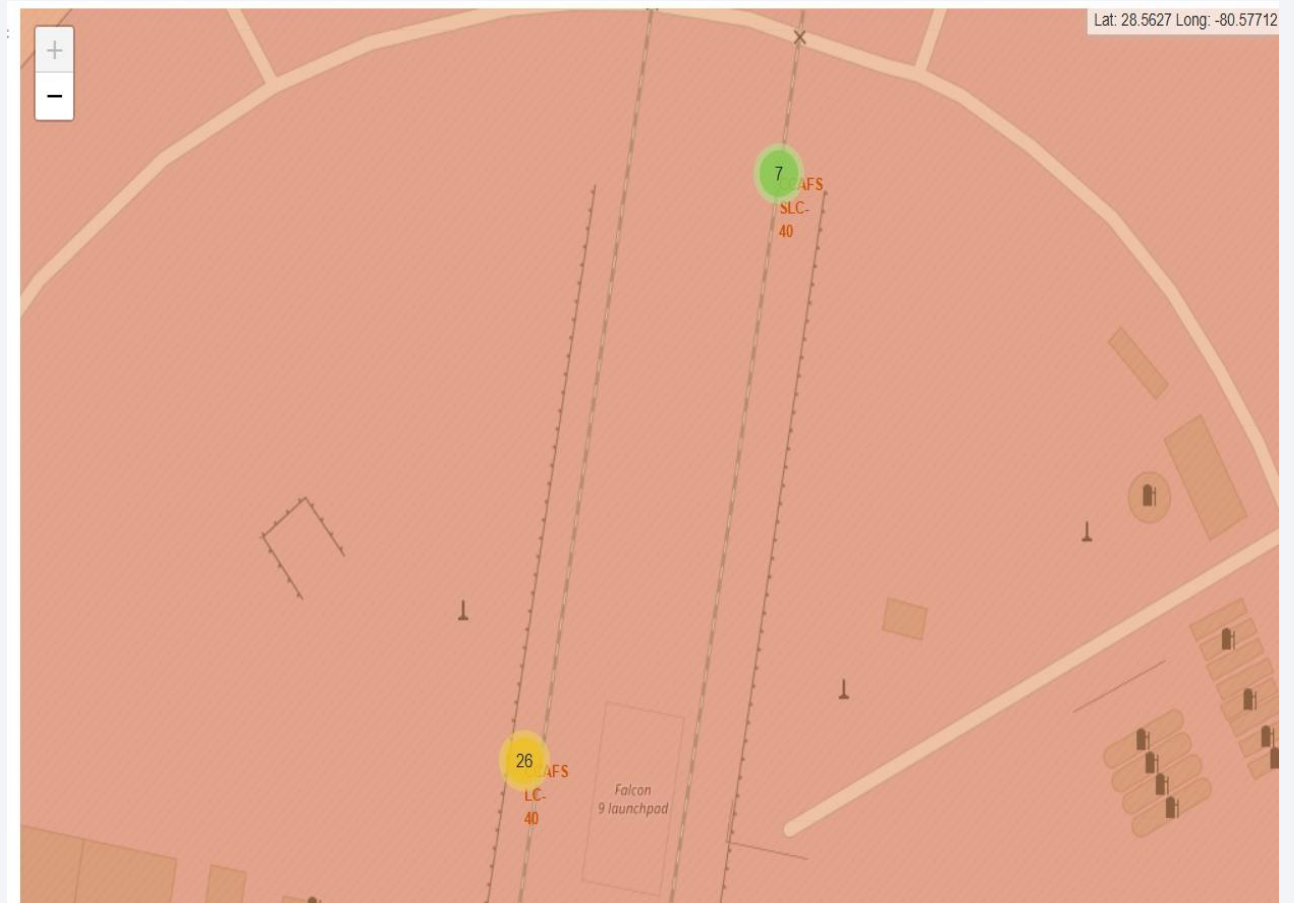
Folium Map Showing the rate of success vs failure at launch sites

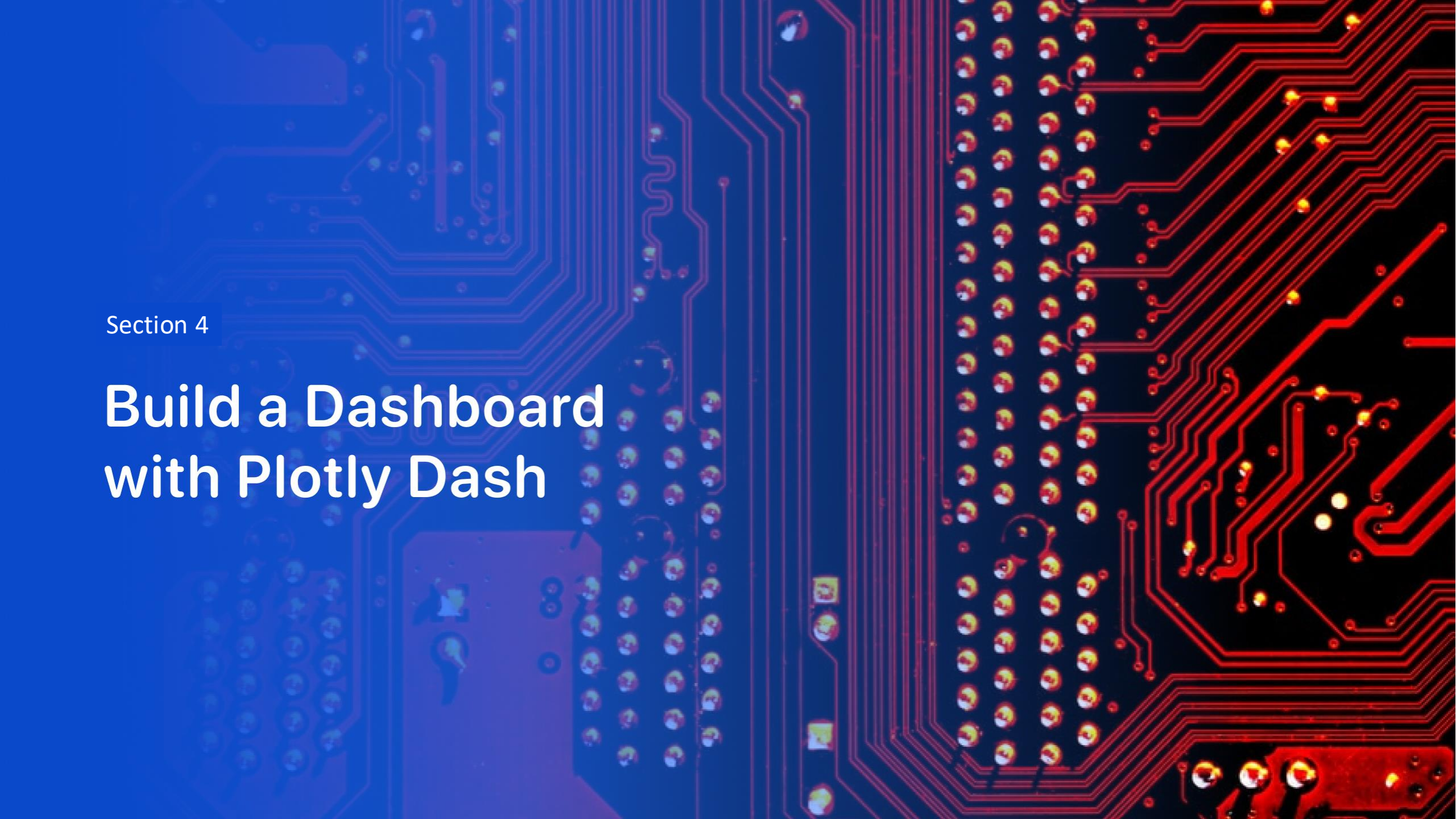
- Colour labelled markers in marker clusters usually represent the success rates of spacex, falcon nine launch sites.
- Sites with relatively high success rates are marked with distinct colours, making it easy to identify them at a glance.
- Three visual cues helping quickly analysing performance trends across different ions, aiding in decision making and further exploration.



Folium Map Showing Distance between launch sites

- Launch sites are generally not in close proximity to railway stations as rail access is not a critical factor for rocket launches.
- Most lawn sites are near highways to facilitate transportation of rocket components and personnel.
- All launch sites are very close to coastlines ensuring safety during launches by directing rockets over the ocean.





Section 4

Build a Dashboard with Plotly Dash

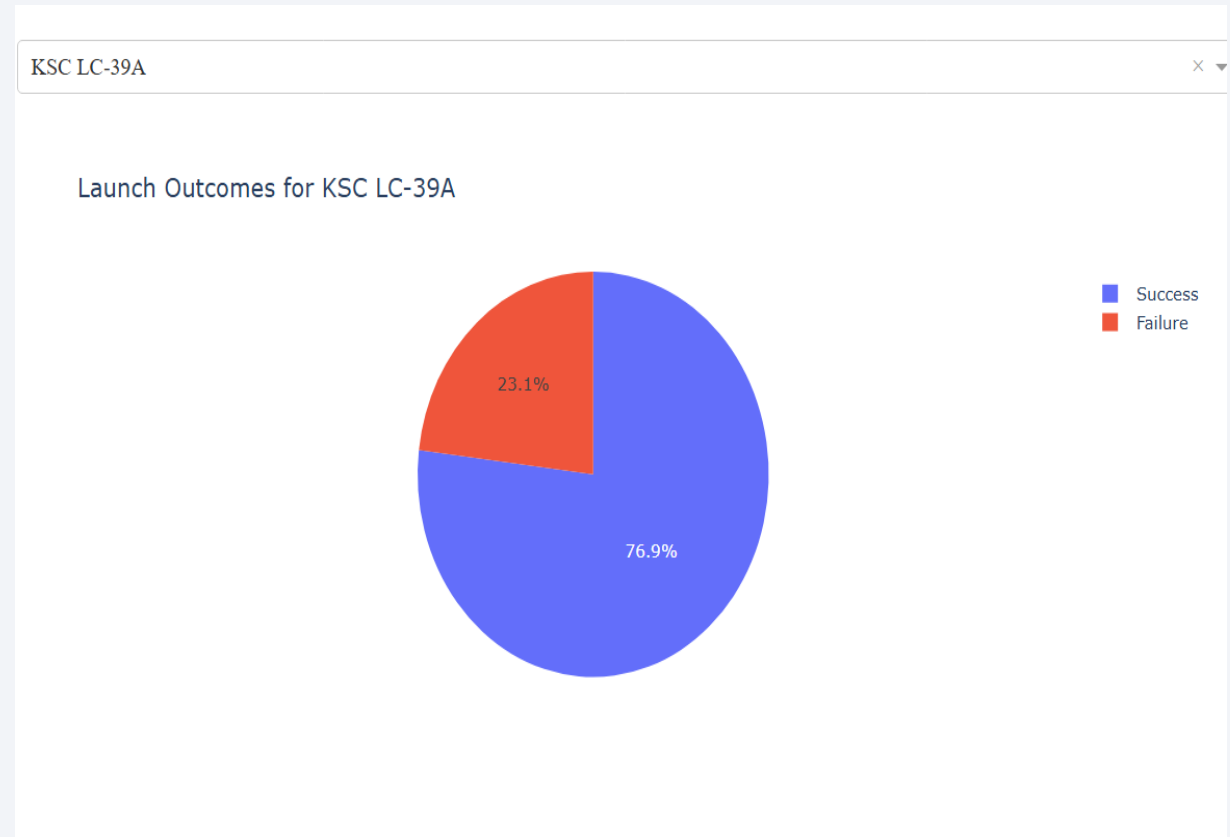
Pie chart of total success launches by launch sites

- Despite chart shows the total number of successful launches by individual launch sites.
- We can clearly see the percentage of each sites contribution for the successful launches.
- KSC LC – 39A has more success rate than the remaining launch sites.



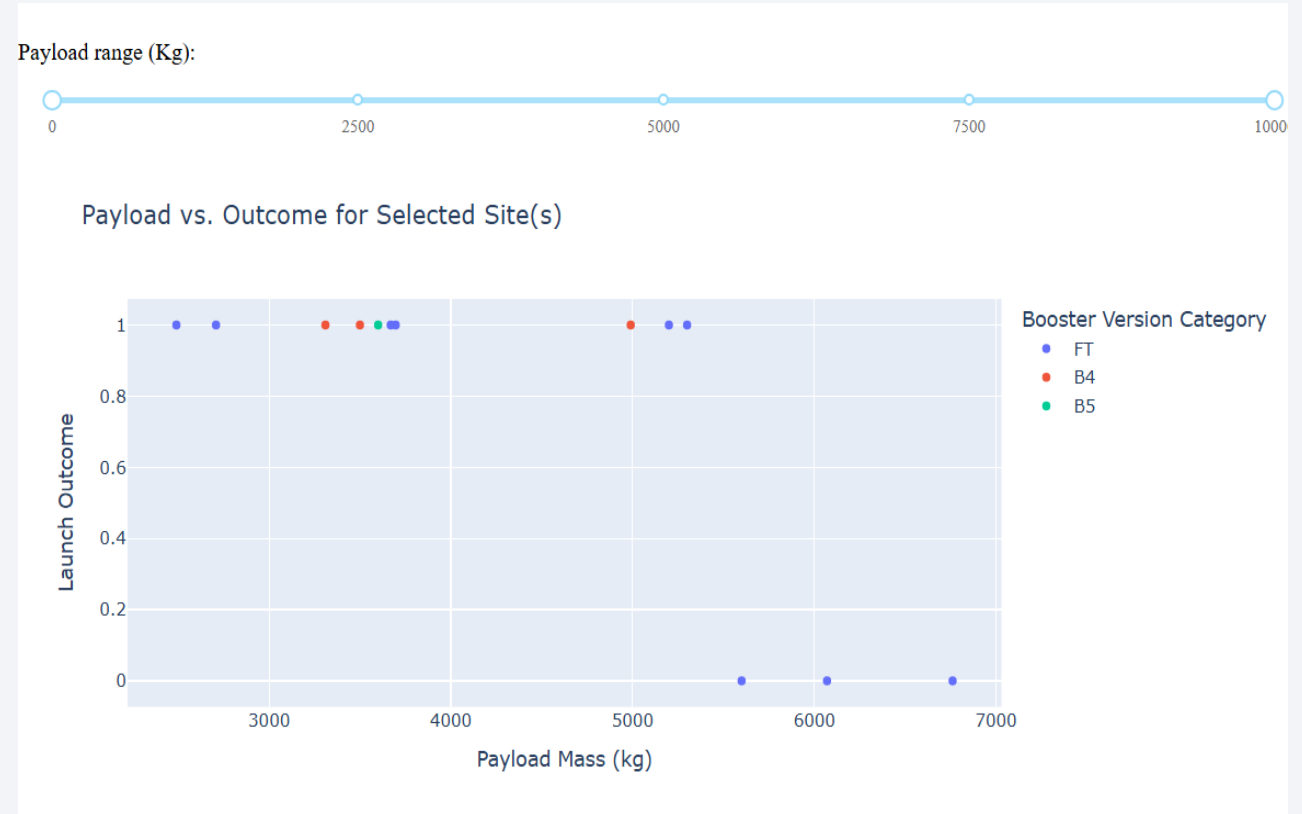
Pie chart for the launch site with highest launch success ratio

- Here is the pie chart of the launch site with highest launch success ratio, which is KSC LC 39A.
- Out of 100 percent, it has 76.99% of success rate with just two, 3.1% of failure rate



Payload vs launch outcomes scatter plot for all launches sites

- The X axis represents the payload mass ranging from zero to 10,000 kg This helps analyse how payload size impacts launch outcomes.
- The plot highlights clusters of successful launches within specific payload ranges suggesting optimal payload limits for higher success rates.
- By filtering for selected sites, the scatter plot provides targeted analysis, showing how payload success varies across different launch locations.





Section 5

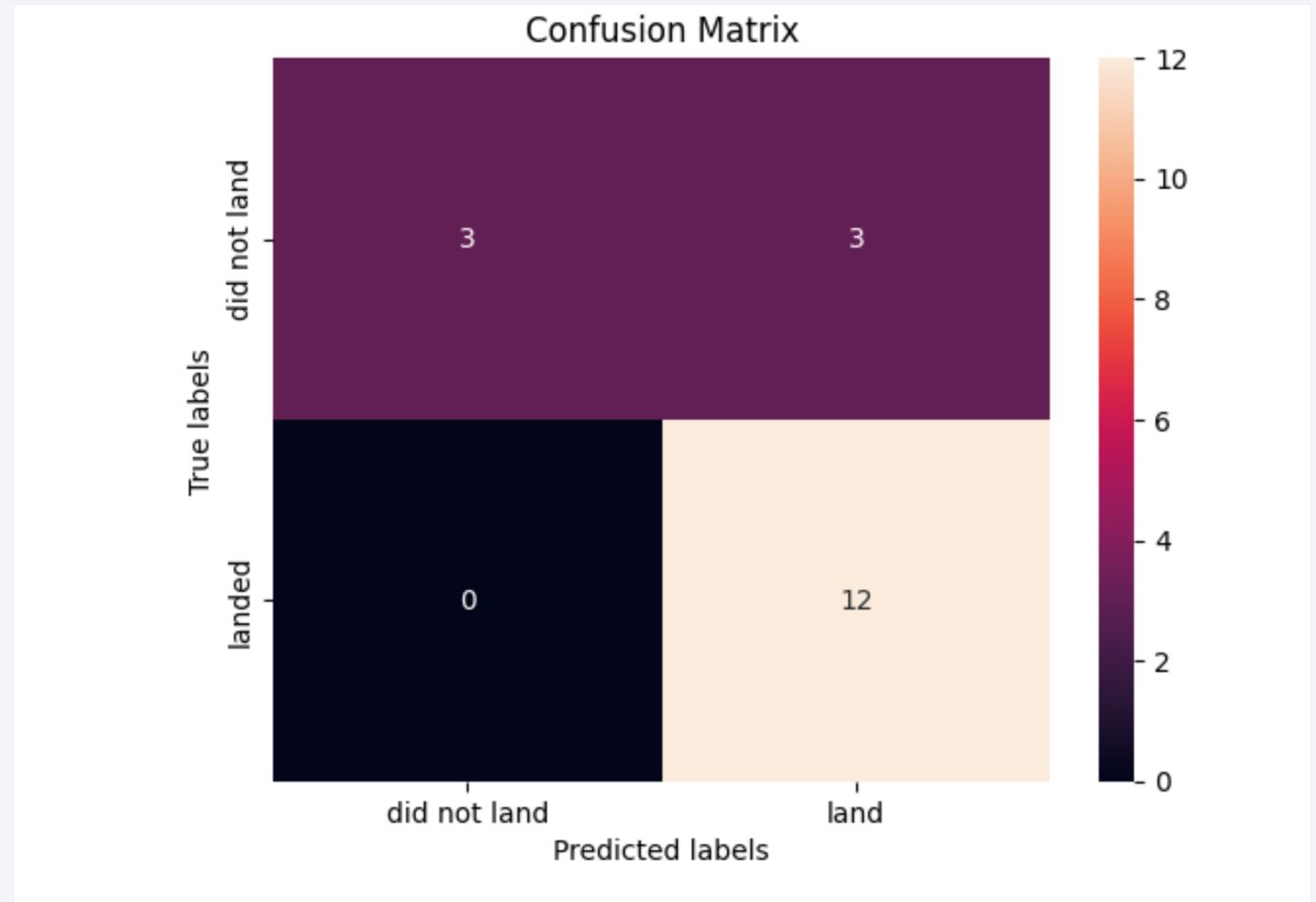
Predictive Analysis (Classification)

Classification Accuracy

- From all the testing models, the random forest classifier emerged as the best performing model.
- It achieved the highest classification accuracy among all tested models demonstrating its reliability.
- Random forest effectively utilised multiple features such as payload mass and launch site to make accurate predictions.
- The model's performance was enhanced through the techniques like grid search, optimising parameters for better results.
- The model is ideal for predicting falcon 9 first stage landing success, aiding in mission planning and decision making.

Confusion Matrix

- So this represents the confusion matrix of the best performing model.
- True positive represent cases where the model correctly predicts successful rocket landings, validating its reliability and true negative cases where field landings are accurately predicted, demonstrating the models precision in identifying failures.



Conclusions

- Successfully collected cleaned and analysed falcon 9 launch data uncovering key insights into factors influencing launch outcomes.
- Developed interactive dashboards and maps using plotly dash and folium, enabling dynamic exploration of launch data and spatial relationships.
- Built and evaluated multiple classification models optimising hyperparameters to identify the best performing model for predicting launch success.
- The project provides actionable insights for improving machine planning, optimising payload configurations and enhancing launch success rates.
- The methodology can be extended to analyse other rocket programmes or incorporate additional data for further refinement.

Thank you!

